Binary Classification of AOI Alarms into True Defects and False Calls

Second check of AOI results with supervised machine learning (Siemens production data)

P2: core code and baseline simulation · test plan · report

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Team 1 · Guided by Prof. Dr.-Ing. Hamidreza Heidari · 3 November 2026 · github.com/bodhi584/aoi-false-call-classification

Yu Han

All three P2 deliverables are in the repository

P2 requirement (lecture)	Status	Where
Core code + baseline simulation	done	aoi_classification/, scripts/replicate_pfab2025.py
Test plan	done	docs/test_plan.md
Report (bibliography report)	done, 31 sources	docs/literature_report.pdf

What changed since P1

P1: build a second check after AOI and test it in time order.

Literature review: on the same Siemens data, such a check already fails on later production days (Pfab & Rothering 2025).

They name two causes: drift and an unstable threshold method.

P2 question: can retraining (against drift) and a correctly set threshold keep the second check working over time?

Method: repeat their experiment one-to-one and change one factor at a time.

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Melisa Subasi

AOI false calls are a real and current problem

≈ 99 %

of the solder joints flagged by AOI are false calls

Each flagged board is checked again by hand. Source: Pfab & Rothering (2025); data: Pfab et al. (2024), one Siemens SMT line, 132 days

Zero-defect manufacturing needs detection [1]

AOI checks every solder joint with a camera and fixed limits [2]; machine learning at the inspection step is an active field [3].

Manual workload [5]

Every flagged board goes to the manual inspection station; many are good boards.

Siemens and NVIDIA, Erlangen, Oct 2026 [4]

A second inspection step decides: real defect or false call. Target: 40 % fewer false calls (a target, not a result).

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Meron Hagos

A second check with machine learning works on a test set

Study	Data	Classifiers	Result
Thielen et al. 2020 [6]	not public	kNN, neural network, random forest	false calls recognised; RF and NN best on large data
Albayari et al. 2026 [7]	Siemens AG	XGBoost, RF, DNN, kNN	XGBoost best balance of recall and precision
Pfab & Rothering 2025 [8]	Siemens, public	kNN, RF, balanced RF, XGBoost, AutoML	RF and balanced RF meet the targets on the test set

Which classifiers? [9]–[14]

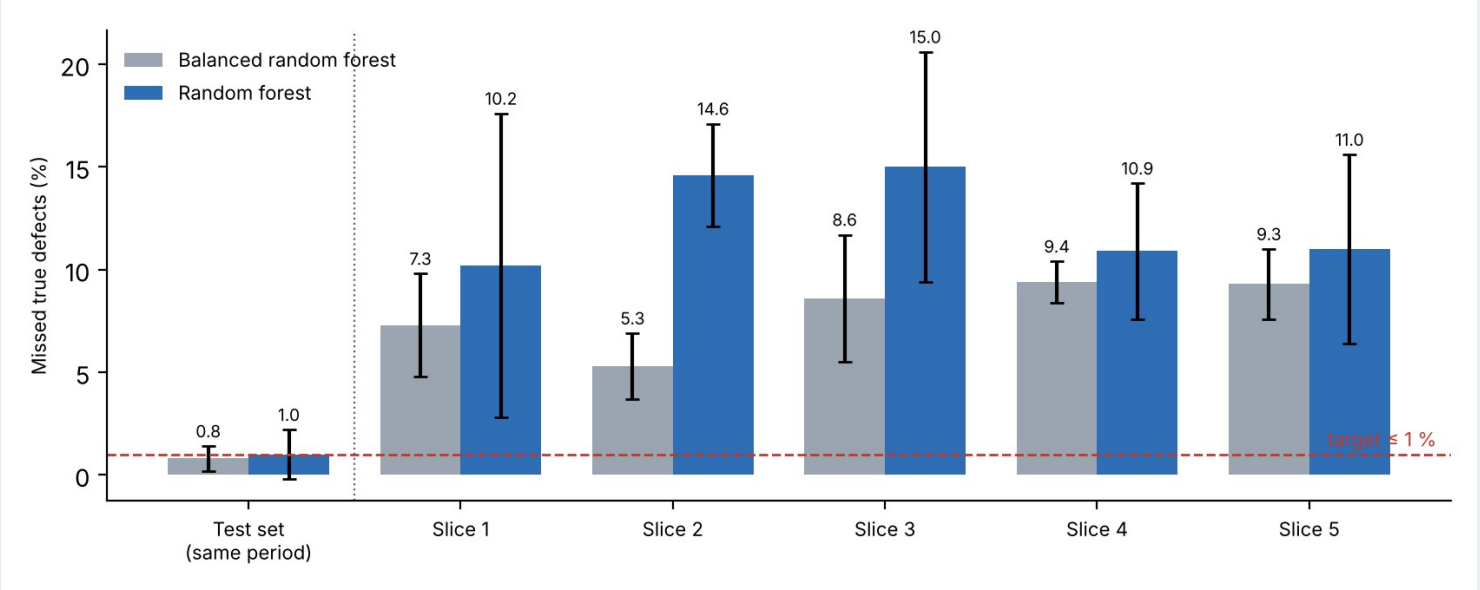
On tabular data, tree ensembles (random forest, gradient boosting) are the strongest family in five benchmark studies.

But: tested on data from the same period

None of these studies keeps the model running on later production days, except [8].

Alejandro Rivas

… but fails on later production days: drift



Why? Drift [22]–[28]

Pfab & Rothering: on target on the test set, 5–15 % missed defects on all five later slices.

Drift = the data change over time; a model trained on old data no longer fits [22]–[24].

Same effect in manufacturing: welding, semiconductor etch, sensors [25]–[27].

The paper also names an unstable threshold method.

Yu Han

Chain of evidence

Fails over time

train once → 5–15 % missed defects later [8]

Two causes

drift [22]–[28] and an unstable threshold method [8]

Remedies

periodic retraining [23], [24], [28]; re-set the threshold when class shares change [21]

Works elsewhere

fraud detection, <1 % positives [30]; semiconductor materials line [31]

Open for AOI

not tested for the second check of AOI results

Our research question

Can periodic retraining and a correctly set threshold keep the second check of AOI results within ≤ 1 % missed true defects and ≥ 40 % fewer manual checks on later production days?

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Same system, sharper question and criteria

INPUT

AOI alarm: AOI measurements of each flagged solder joint

SYSTEM

Binary classifier after the AOI; retrained and threshold re-set during production

OUTPUT

True defect → manual inspection · false call → released

	P1 (23 Oct)	P2 (now)
Question	build a second check and test it in time order	can retraining and a correct threshold keep it working over time?
Criteria	board level: recall ≥ 99 %, specificity ≥ 20 %	joint level, as in [8]: missed defects ≤ 1 %, false calls released ≥ 40 %
Why	own targets	one-to-one comparison with [8]

Yu Han

Replication of Pfab & Rothering (2025): one factor at a time

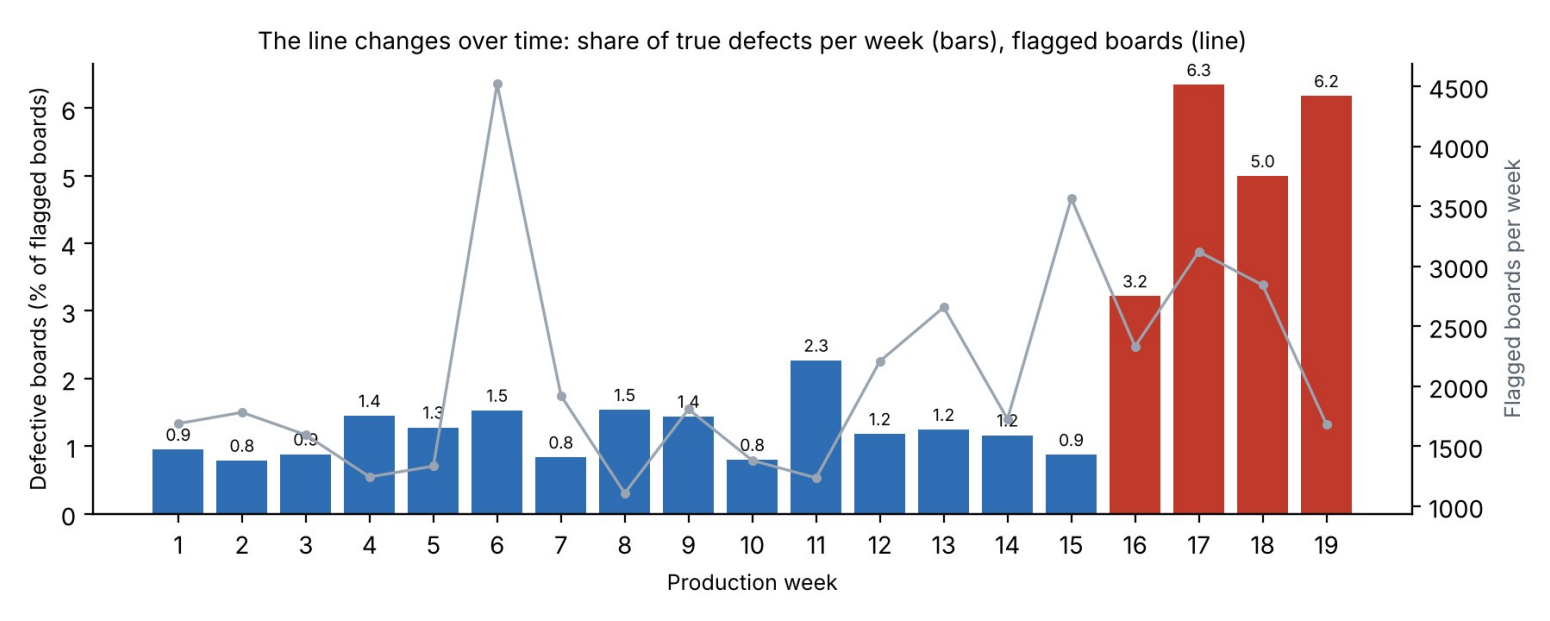
	(a) Paper exactly	(b) Train once, threshold re-estimated	(c) Retrain
Data split	same as [8]	same	same
Models, hyper-parameters, seeds	RF, balanced RF; logged values; 10 seeds	same	same
Threshold	value logged in [8]	their method, applied to the final model	same method, new data
Training	once	once	before each of the 5 later slices
Purpose	reproduce [8]	(a) vs (b) = effect of the threshold	(b) vs (c) = effect of retraining

Metrics on joint level, as in [8]: missed true defects (slip = 1 − recall) and false calls released (volume reduction = specificity). Assumption: manual-inspection labels are available for all joints (upper bound for retraining).

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Alejandro Rivas

The line changes over time



Facts (our analysis)

440,274 flagged joints, 39,742 boards, 132 days

97.8 % of flagged boards are false calls

Share of defective boards: 0.8–2.3 % per week, then 3.2–6.3 % in the last 4 weeks

Drift test: early vs. late days separable, AUROC 0.974

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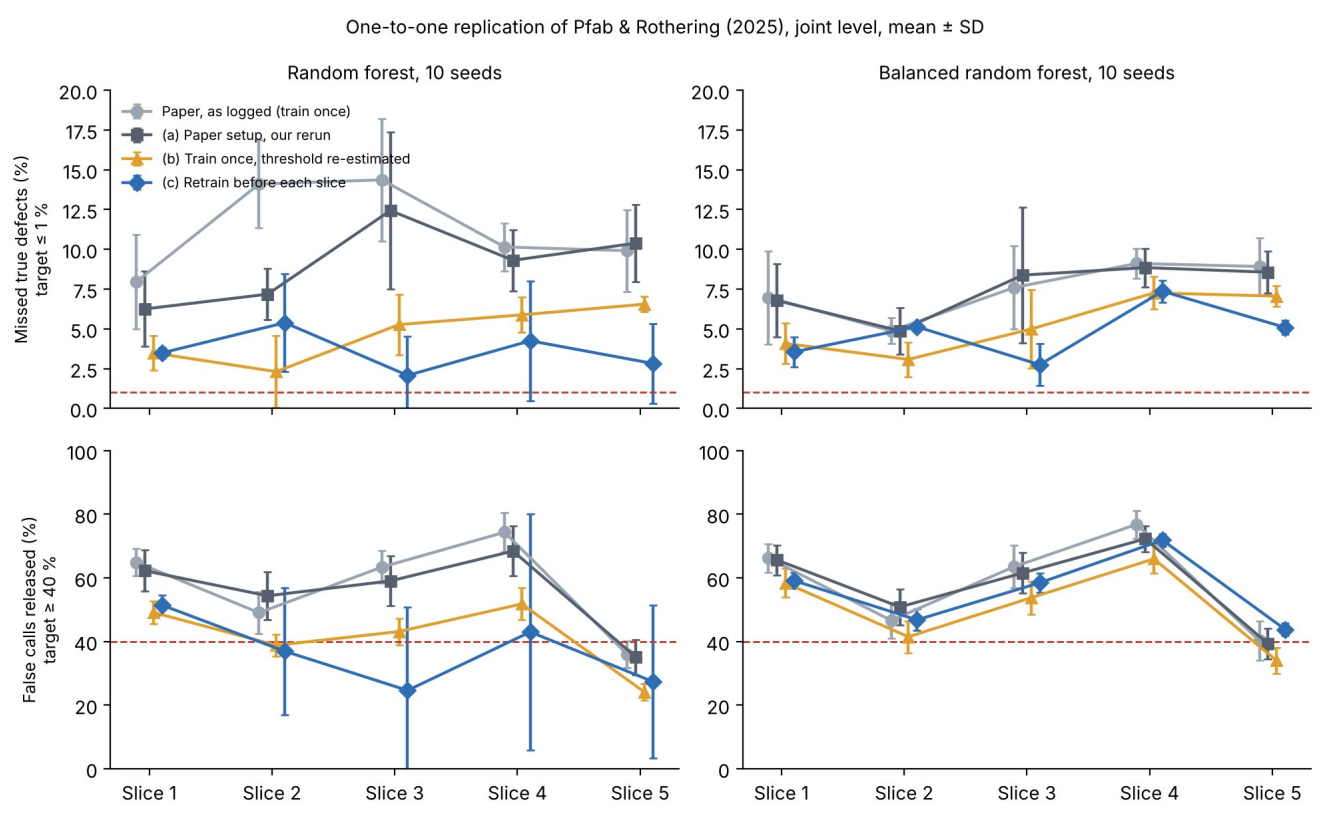
What runs in the repository

Part	File	What it does
Data	aoi_classification/data.py	load the Siemens data, boards, time order
Classifier + threshold	aoi_classification/model.py	train, score, threshold for 99 % recall
Baseline simulation	aoi_classification/simulation.py	day by day: train on earlier days only
Replication	scripts/replicate_pfab2025.py	one-to-one repeat of [8], train once vs. retrain
Classifier screening	scripts/compare_classifiers.py	13 classifiers, 8 families, same split
Tests + CI	tests/, GitHub Actions	10 tests on Windows, macOS, Linux

Tools: Python, pandas, scikit-learn, imbalanced-learn, matplotlib, Jupyter · Data: Mendeley (CC BY-NC 3.0), not stored in the repo

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Replication works; retraining helps a little, target not reached yet



Reproduction ✔

Balanced RF: 7.5 % missed defects, as in [8].

Retraining: small gain

Balanced RF: 5.3 → 4.8 % missed, 50.7 → 56.0 % released.

RF: 4.7 → 3.6 % missed, but fewer released.

Threshold matters most

Correct threshold alone: 7.5–9.1 → 4.7–5.3 % missed.

No variant reaches ≤ 1 % yet.

Melisa Subasi

How we check the code and the numbers

Group	Examples	Status
Data (D1–D7)	labels 0/1, no missing values, time order, checksum	4 exist, rest P3
Classifier + threshold (C1–C7)	threshold rule, all models run, same seed = same result	4 exist, rest P3
Simulation (S1–S4)	model only knows earlier days	1 exists, rest P3
Replication (R1–R4)	same split as [8]; (a) reproduces [8]; retraining uses only earlier data; 10 seeds	new
KPIs (K1–K5)	missed defects ≤ 1 %, false calls released ≥ 40 %; every number from results/	open

Unit tests run on every push (CI, three operating systems). A KPI that is not reached is a result, not a failed test.

Meron Hagos

Towards P3 (24 Nov)

Retraining frequency

daily, every 3 days, weekly, per slice: how far do missed defects drop? [31]

Threshold on recent data

set the threshold on the latest days, not on all history [21]

More model families

gradient boosting, neural network, logistic regression in the same setup

Simulation tool

one screen: choose the retraining frequency, see missed defects and saved checks

KEY REFERENCES

Full list (31 sources) in docs/literature_report.pdf

[5] Pfab, K. et al. (2024). Data of automated optical inspection of surface-mounted technology electronic production. Data in Brief 53, 110110.

[8] Pfab, K., Rothering, M. (2025). Towards improved research methodologies for industrial AI: a case study of false call reduction. arXiv:2506.14521.

[22] Moreno-Torres, J. G. et al. (2012). A unifying view on dataset shift in classification. Pattern Recognition 45, 521–530.

[23] Gama, J. et al. (2014). A survey on concept drift adaptation. ACM Computing Surveys 46(4), 44.

[24] Lu, J. et al. (2019). Learning under concept drift: a review. IEEE TKDE 31(12), 2346–2363.

[30] Dal Pozzolo, A. et al. (2018). Credit card fraud detection: a realistic modeling and a novel learning strategy. IEEE TNNLS 29(8), 3784–3797.

[31] Gao, M. et al. (2026). Robust and reliable AI for predictive quality in semiconductor materials manufacturing with MLOps and uncertainty quantification. arXiv:2605.07752.

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Repository

github.com/bodhi584/aoi-false-call-classification

Thank you

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